

I. Fluorescence-activated cell sorting for single cell genomics

General

FACS sorting and whole genome amplification is performed in a cleanroom. Protective clothing such as lab coats, shoe covers, are required for entering the room and additional protection with gloves, hair nets and mouth covers are needed when the plates are sorted into and further processed. Lab coats should be dedicated to a specific task/analysis i.e. sorting or whole genome amplification. The screening of the amplified material is done outside the cleanroom in a PCR hood dedicated to DNA work. To keep a clean environment, take off your gloves when using the computer or touch screen. For a more detailed description of the different steps and critical issues see the paper by Rinke et al 2014 (1).

Preparations

Sort plate preparation

- 1 µl of sterile filtered 1x PBS is pipetted to each well of a 96-well plate. The plate is covered by a clear sealing and then UV-treated for 2 Joule.
- Prepared plates are kept at 4°C until sorting the same day.

Clean single cell sort MoFlo Astrios EQ

- Sterile filter five liters of 1 x PBS through a 0.1 µm top filter, and pour it into the sheath tank. The fluid is left about 30 min to let air bubbles dissolve. During this time the MoFlo Astrios EQ cell sorter is run only with ddH₂O as sheath. Also run water through the sample line for some time.
- A coarse stream alignment and fine alignment of the lasers are performed using 3.08 µm SPHERO Ultra Rainbow Fluorescent Particles until the adjustments pass the QC.
- The CyClon moving arm is adjusted in the plate protocol in accordance to the center of the wells on the plate.
- The manual drop delay is adjusted using 10 µm Flowcheck beads until satisfaction.
- Single cell sorting is done with the single drop sort mode and 0.5 drop envelope.

Protocol

Staining of sample

- Thaw the sample on ice.
- Run the sample through a 70 µm cell strainer to get rid of cell aggregates.
- Run the unstained sample for sample analysis.
- Depending on the event rate (eps = events per second), the sample is diluted accordingly.
- Add SYBR Green I to a final concentration of 1x to the sample.
- Stain the sample in the dark and on ice for at least 20 min.
- During analysis of the stained samples the region of interest for sorting are chosen.

- Define the sort layout in the CyClone plate layout. Always include a few empty wells as negative controls and a few wells with more than 1 cell (e.g 10 cells) as positive controls.
- Clean the sort chamber and the work space with a decontamination solution, e.g. DNA AWAY®.

Running a clean single cell sort

- Take one plate, carefully remove the sealing.
- Place the plate into the sort chamber, pay attention that the plate is positioned correctly.
- Start the sample, then start the sort.
- Once the sort is finished, carefully take out the plate and immediately cover it with a sealing. Spin down, and transfer the plate to the freezer.
- Continue with the next plate.
- Sorted plates are kept at -20°C for short-term, or -80°C for long-term storage.

Cleaning procedure

- Take out the sample.
- Run several backflushes.
- Run bleach in the sample line with increased sample pressure for at least 5 minutes.
- Run cleaning fluid in the sample line.
- Run water in the sample line.

II. Cell lysis and whole genome amplification using MDA

Whole genome amplification is performed as described for the “amplification of genomic DNA from single cells” in the REPLI-g Single Cell Handbook (2), with the exception that volumes are scaled down with a quarter of the original volume, hence 12.5 µl in total.

Lysis and neutralization reagents from the REPLI-g Single Cell kit are first added to each well of the sorted plate to facilitate genome amplification. The MDA master mix is then added, the plate is sealed with an optical sealing and the reaction is monitored every 15 minutes in real-time at 30 °C for 8 h using a DNA-binding dye (SYTO13) and an instrument suitable for fluorescent readings e.g. a PCR machine or a plate reader. Note that the REPLI-g Single Cell kit mastermix should not be UV-decontaminated as it is already treated to minimize contamination. Tubes, water, lysis and stop solution can however be UVed for 2 Joules before use.

General cleanroom rules and cleaning procedure

- Always wear shoe covers when entering the cleanroom.
- Use a dedicated lab coat, and only wear it inside the cleanroom, use it at all times.
- Use hair net and mouth cover when you set up the MDA reaction.

- Use fresh gloves each time you re-enter the cleanroom, use gloves for every single step.
- Wipe all working areas and pipets with DNA AWAY®.
- Clean your gloves and everything that goes into the MDA hood with DNA AWAY®.

General considerations

To minimize contamination

- Change tips between every well (or rows/columns if you use a multichannel pipette).
- Mix and spin down all reagents before use to avoid possible drops in the lid coming in contact with your gloves.
- Open sorted plates only in the PCR hood.
- Include negative controls such as “no sort controls” and no template controls if possible, to keep track of potential contamination.

Reagent handling

- Keep all reagents on ice after thawing or preparation. The polymerase should be kept in a cooling block.
- After adding the nucleic binding dye SYTO13, protect the MDA mastermix from light using foil.

Preparation for MDA

- Clean the hood working area with DNA AWAY®.
- Wipe pipettes with DNA AWAY®.
- Wipe all equipment that you will need, including enough pipette tips, with DNA AWAY® and place them in the hood.
- Turn on the UV in the hood for 20 to 30 min.
- Fill an ice-box with ice and place a lid wrapped in foil on top of it, fill this lid with ice (will be used for the UV-linker).
- Fill an ice-box with ice for the reagents later on.

Protocol

Prepare the lysis, stop solution and MDA master mix

- Thaw plates on ice in an ice box
- Add lysis buffer (Buffer D2) in an 8-well strip and close with an 8-cap lid. Do the same for the stop solution.
- Mix all reagents for the MDA, including the SYTO13 dye, and distribute it in an 8-well strip and close with an 8-cap lid. Protect the reaction from light.

Lysis and neutralization

- Add 0,75 µl lysis reagents on the rim of the wells in each row using a multichannel pipette.
- Close the plate with a sealing and spin down.
- Put the plate in a thermocycler at 65° C for 10 min.
- Add 0,75 µl stop solution on the rim of the in each row using a multichannel pipette.
- Close the plate with sealing and spin down.

MDA

- Add 10 µl of MDA-mix reagents on the rim of the wells in each row using a multichannel pipette.
- Close the plate with a sealing and spin down.
- Put the plate in a qPCR machine or a plate reader and incubate the plate at 30 °C for 8 h while reading the fluorescence every 15 minutes.
- The reaction is inactivated at 3 min 65 °C

Screening of amplified material

Screening of whole genome amplification (WGA) products is done using the ROCHE lightCycler 480 and the ROCHE kit LightCycler 480 SYBR Green I Master in 10 µl reactions. Dilute PCR products x 100 and add 2 µl of the DNA to the PCR-mix including the forward and reverse primer. Add a melting curve analysis after the PCR cycling for evaluation of positive reactions by comparing T_m-values of the samples and positive as well as negative controls.

Reagents	final conc	x1 (µl)
H2O		2,5
Fwd primer 10 µM	0.25 µM	0,25
Rev primer 10 µM	0.25 µM	0,25
SYBR I Mastermix x 2		5
Amplified DNA		2
		10

Purification of PCR-products

Purified PCR-products or WGA can be quantified in a plate format using the Quant-IT PicoGreen® dsDNA Assay kit (Thermo Fisher Scientific) and a plate reader as described in the REPLI-g Single Cell handbook from Qiagen Appendix B (2).

References

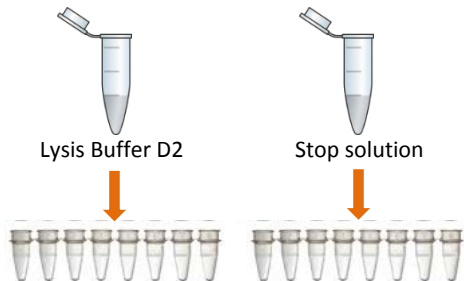
(1). Rinke C, Lee J, Nath N, Goudeau D, Thompson B, Poulton N, Dmitrieff E, Malmstrom R, Stepanauskas R, Woyke T. Nat Protoc. 2014 May;9(5):1038-48. doi: 10.1038/nprot.2014.067. Epub 2014 Apr 10. PMID: 24722403

(2). REPLI-g® Single Cell Handbook, Sample & Assay Technologies, Qiagen 2012.

Overview of MDA

Lysis and neutralisation

REPLI-g Single Cell kit buffers



	x1 (μl)
Cell + sort buffer	1
Lysis buffer	0,75
Stop solution	0,75
MDA Master-mix	10
	12,5

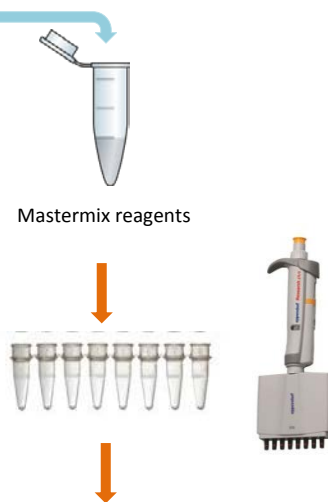
Alkaline lysis

- Add 0,75 μl lysis reagents on the rim of the well using a multichannel pipette
- Close the plate with sealing and spin down
- Put plate in thermocycler at 65° C for 10 min
- Add 0,75 μl lysis reagents on the rim of the well using a multichannel pipette
- Close the plate with sealing and spin down

MDA

REPLI-g Single Cell kit

Reagent	x1
H2O	2,22
REPLI-g sc reaction buffer	7,25
REPLI-g sc DNA pol	0,50
Syto 13 200 μM	0,03
DNA +lysis/stop	2,50
Mix	12,50



Procedure

- Prepare the mastermix
- Add the mix to each well in an 8-well strip
- Add 10 μl MDA-mix to each row using a multichannel pipette
- Change tips between every row
- Close the plate with a sealing and spin down
- Incubate the plate at 30 °C for 8 h
- Inactivate for 3 min at 65 °C

	1	2	3	4	5	6	7	8	9	10	11	12
A	1	1	1	1	1	1	1	1	1	1	1	1
B	1	1	1	1	1	1	1	1	1	1	1	1
C	1	1	1	1	1	1	1	1	1	1	1	1
D	1	1	1	1	1	1	1	1	1	1	1	1
E	1	1	1	1	1	1	1	1	1	1	1	1
F	1	1	1	1	1	1	1	1	1	1	1	1
G	1	1	1	1	1	1	1	1	1	1	1	1
H	1	1	1	1	1	1	1	1	1	1	1	1